

Difference Between JPA, Hibernate, and Spring Data JPA

JPA (Java Persistence API)

JPA is a specification that defines standard rules for mapping Java objects to database tables. It provides annotations like `@Entity`, `@Table`, `@Id`, and `@GeneratedValue`. JPA does not perform database operations by itself; it requires an implementation such as Hibernate.

Example:

```
@Entity
@Table(name = "students")
public class Student {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String name;
}
```

Hibernate

Hibernate is an ORM (Object Relational Mapping) framework that implements the JPA specification. It performs database operations such as insert, update, delete, and fetch records. Hibernate automatically converts Java objects into database records and generates SQL queries, reducing the need to write SQL manually.

Example:

```
Student student = new Student();
student.setName("Ravi Kumar");

session.save(student); // Hibernate generates the INSERT query
```

Spring Data JPA

Spring Data JPA is a Spring Framework module built on top of JPA. It simplifies database programming by providing ready-made repository interfaces like `JpaRepository`. Developers can perform CRUD operations using methods such as `save()`, `findAll()`, `findById()`, and `deleteById()` without writing DAO implementations.

Example:

```
public interface StudentRepository extends JpaRepository<Student, Long> {
}

studentRepository.save(student);
studentRepository.findAll();
```

Relationship

JPA defines the standard rules, Hibernate implements those rules and communicates with the database, and Spring Data JPA makes working with JPA much simpler by reducing boilerplate code.

In simple words:

- JPA : Defines the rules (Specification).
- Hibernate : Implements the rules and communicates with the database.
- Spring Data JPA : Simplifies database operations by providing built-in repository methods.

Conclusion

All three technologies work together. JPA provides the standard, Hibernate performs the actual database operations, and Spring Data JPA makes development faster and easier by reducing the amount of code developers need to write.